DATA ANALYTICS PORTFOLIO PROJECT

Supply Chain Performance & Inventory Analysis

A practical review of inventory, demand, suppliers, manufacturing, logistics and product quality

PROJECT DETAIL	INFORMATION
Prepared by	Darshana Narjari
Dataset	Supply Chain Analysis
Dataset source	Harsh Singh – Kaggle
Tools	PostgreSQL • Microsoft Power BI • CSV
Analysis grain	One record per SKU in the supplied snapshot
Project type	Data Analytics Portfolio Project

PROJECT REPORT | 2026

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Project at a glance

This report uses SQL to validate and analyse a 100-row supply chain snapshot, then uses Power BI to present the results in three dashboard pages. It is written as a descriptive analysis: the purpose is to identify patterns and areas for further investigation, not to claim that the dataset can forecast demand or automatically optimise operations.

1. Executive Summary

This project reviews supply chain performance across 100 unique SKUs in three product types: skincare, haircare and cosmetics. The analysis compares units sold with stock on hand, reviews shipping time and cost by transportation mode, examines inspection outcomes and defect rates, and compares supplier and manufacturing measures.

PostgreSQL was used to create a raw table, run data-quality checks and prepare grouped analysis queries. Microsoft Power BI was used to present the findings in three pages: Inventory Optimization Overview, Logistics & Quality Performance, and Supplier & Manufacturing Performance.

MEASURE	RESULT	INTERPRETATION
SKUs	100	100 unique SKU records
Units sold	46,099	Sum of Number of products sold
Stock on hand	4,777	Sum of Stock levels
Revenue generated	577,604.82	Currency is not specified in the source file
Average shipping time	5.75	Average of Shipping times; unit not specified
Average defect rate	2.28	Average of Defect rates; scale not specified

Key messages

- Skincare contributes the highest units sold and revenue among the three product types.
- 21 SKUs meet the simple high-demand/low-stock screening rule; 25 meet the high-stock/low-sales rule.
- Road has the shortest average recorded shipping time, while Sea has the lowest average shipping cost.
- Haircare has the highest average defect rate; 36 inspection records are Fail and 41 are Pending.
- Supplier 1 has the lowest average supplier lead time and defect rate in this dataset; Supplier 5 has the highest average defect rate.

These results are screening signals from a single snapshot. They do not confirm stockouts, prove excess inventory, establish causes of defects or predict future demand.

2. Business Context, Problem & Objectives

Business context

A company needs to manage its products, inventory, suppliers, manufacturing and transportation properly. It must ensure that enough products are available to meet customer demand, suppliers deliver on time, products are manufactured efficiently and quality standards are maintained. This project analyses supply chain data to understand how these areas are performing and identify where improvements may be needed.

Business problem

The company needs a better understanding of its inventory, suppliers, manufacturing, transportation and product quality. Without analysing these areas together, it may be difficult to identify products with low stock, delays in delivery, supplier performance issues and quality problems. This project uses data analysis to identify these areas and help the company make better business decisions.

Project objectives

- Understand the structure and quality of the supplied dataset.
- Compare units sold with stock levels to flag possible inventory pressure.
- Compare shipping time and shipping cost across transportation modes.
- Review defect rates and inspection outcomes by product type.
- Compare supplier lead time, defect rate and shipping cost.
- Compare manufacturing lead time and manufacturing cost by product type.
- Present findings in Power BI and recommend practical follow-up actions.

Key business questions

- Which product types account for the most units sold and stock?
- Which SKUs combine relatively high sales with relatively low stock?
- Which transportation mode has the shortest average shipping time?
- How do defect rates differ across skincare, haircare and cosmetics?
- Which suppliers have longer average lead times or higher average defect rates?
- Which product types have higher average manufacturing lead time and cost?

Scope: descriptive analysis of a 100-row snapshot. There are no transaction dates, demand history, purchase-order dates, stockout events, service-level targets, supplier contracts or profit measures in the supplied file.

3. Dataset Overview & Data Dictionary

The project uses the file supply_chain_data.csv. It contains 100 rows and 24 columns. Each row represents one SKU-level record in the supplied snapshot. The data covers three product types, five suppliers, five locations and four transportation modes.

DATASET DETAIL	VALUE
File	supply_chain_data.csv
Rows / columns	100 rows / 24 columns
Unique SKUs	100
Product types	Skincare, Haircare, Cosmetics
Supplier names	5 suppliers
Locations	5 locations
Transportation modes	Road, Rail, Air, Sea

Data dictionary – field groups

FIELD GROUP	COLUMNS	WHAT THEY REPRESENT
Product	Product type, SKU, Price	Product category, identifier and listed price
Sales	Number of products sold, Revenue generated	Recorded unit sales and revenue measure
Inventory	Availability, Stock levels, Order quantities	Availability and inventory-related quantities
Supplier	Supplier name, Location, Lead time, Lead times	Supplier identity, location and two distinct lead-time fields
Manufacturing	Production volumes, Manufacturing lead time, Manufacturing costs	Production and manufacturing measures
Shipping	Shipping times, Shipping carriers, Shipping costs, Transportation modes, Routes	Transportation mode, carrier, route, time and cost
Quality	Inspection results, Defect rates	Inspection status and numerical defect measure
Other	Customer demographics, Costs	Additional customer and cost fields

Important: the file contains two similarly named fields, Lead times and Lead time. The supplier comparison in this report uses Lead time. The fields should not be combined until their definitions are confirmed with the data owner.

4. Tools, Methodology & Data Quality

Tools used

TOOL / Method	ROLE IN THIS PROJECT
CSV	Source file containing the SKU-level records.
PostgreSQL	Create the raw table, inspect data quality and run analysis queries.
Microsoft Power BI	Build interactive report pages using KPI cards, slicers, scatter plots and comparison charts.

Analytical workflow

STEP	ACTIVITY	OUTPUT
1	Understand the data structure	100 records, 24 fields, one row per SKU
2	Create the raw SQL table	supply_chain_raw
3	Run data-quality checks	Missing values, duplicate SKUs, negative values and categories
4	Run analysis queries	Inventory, supplier, manufacturing, shipping and quality summaries
5	Build Power BI dashboards	Three report pages
6	Interpret results	Findings, caveats and recommendations

Data-quality results

CHECK	RESULT	NOTE
Missing values	0 across the uploaded CSV	SQL query checks five key fields; file-level validation checked all 24 columns.
Duplicate SKUs	0	100 unique SKU identifiers.
Negative numeric values	0	No negative values found in numeric fields.
Inspection results	Pass 23; Fail 36; Pending 41	Pending is not equivalent to Pass.

These checks support the structural quality of the supplied file. They do not prove that every value is realistic or that all business definitions are understood. The units for availability, lead time, shipping time, defect rate and cost fields should be confirmed before operational targets are set.

5. Inventory & Demand Analysis

The dataset records 46,099 units sold and 4,777 units in stock across 100 SKUs. Average units sold per SKU is 460.99 and average stock per SKU is 47.77. These simple averages are used as screening thresholds in the SQL queries.

Screening approach

The stock-to-sales ratio is calculated as Stock levels divided by Number of products sold. A NULLIF condition prevents division by zero. High-demand/low-stock candidates are records with units sold above the dataset average and stock below the dataset average. The reverse comparison identifies high-stock/low-sales candidates.

SCREENING GROUP	RULE	RECORDS FLAGGED
High demand / low stock	Units sold above average AND stock below average	21 SKUs
High stock / low sales	Stock above average AND units sold below average	25 SKUs

Interpretation

The 21 products with high sales but low stock should be checked to see if they need more stock. The 25 products with high stock but low sales should be reviewed to understand why they are not selling well and whether more orders are necessary. However, we cannot say that these products are definitely out of stock or have excess stock. They are only products that need further checking.

Recommended follow-up

- Check recent demand and available stock for the 21 high-demand/low-stock candidates.
- Check how long new stock will take to arrive and whether an order is already on the way before deciding to rush another delivery.
- Review the 25 high-stock/low-sales candidates before placing additional orders.
- The company should decide when to order new stock by considering how much of a product customers will buy while waiting for the supplier to deliver, plus some extra stock kept as a backup.

A reliable reorder decision also requires demand over time, replenishment history, stockout history, minimum order quantities and service-level targets. Those fields are not available in this snapshot.

6. Product-Type Performance

Product-type comparisons show how SKU counts, unit sales, stock and revenue are distributed across skincare, haircare and cosmetics.

PRODUCT TYPE	SKUs	UNITS SOLD	STOCK	REVENUE GENERATED
Skincare	40	20,731	1,608	241,628.16
Haircare	34	13,611	1,644	174,455.39
Cosmetics	26	11,757	1,525	161,521.27

What stands out

- Skincare has the highest number of SKUs, units sold and revenue in the supplied data.
- Haircare has the highest aggregate stock level of the three product types.
- Cosmetics has the lowest units sold and revenue total.

Revenue is the sum of the supplied Revenue generated field. The CSV does not specify a currency, so the figures are shown without a currency symbol. These totals describe this dataset only; they do not indicate profit because the file does not provide a verified product-level profit measure.

Business use

The company should first identify which product category needs more attention. Then, it should check individual products, their available stock, sales and supplier delivery time before making inventory decisions.

7. Logistics & Transportation Analysis

Shipping performance is compared across four transportation modes using the average recorded shipping time and average shipping cost for each mode.

MODE	RECORDS	AVG. SHIPPING TIME	AVG. SHIPPING COST
Road	29	4.72	5.54
Air	26	5.12	6.02
Rail	28	6.57	5.47
Sea	17	7.12	4.97

Key observations

- Road has the shortest average shipping time at 4.72.
- Sea has the longest average shipping time at 7.12.
- Sea has the lowest average shipping cost at 4.97.
- Air has the highest average shipping cost at 6.02.

Interpretation and action

No transportation mode is automatically best for every shipment. A faster mode may be suitable for urgent or high-priority products, while a lower-cost mode may suit less time-sensitive shipments. The business should compare cost and time together with delivery promises, shipment distance, product value, damage risk and customer expectations.

Before making a carrier-allocation decision, confirm the unit of Shipping times and whether Shipping costs is recorded per unit, shipment or SKU record. Route-level and carrier-level comparisons would provide a stronger basis for action.

8. Product Quality & Defect Analysis

The dataset contains an inspection result for each SKU record and a numerical Defect rates field. This section reviews the distribution of inspection outcomes and compares average defect rates by product type.

INSPECTION RESULT	RECORDS	SHARE OF RECORDS
Pass	23	23.0%
Fail	36	36.0%
Pending	41	41.0%

PRODUCT TYPE	AVERAGE DEFECT RATE
Haircare	2.48
Skincare	2.33
Cosmetics	1.92

Key observations

- Haircare has the highest average recorded defect rate at 2.48.
- Cosmetics has the lowest average recorded defect rate at 1.92.
- 36 records are marked Fail and 41 are Pending; these should not be treated as approved products without a defined release process.

Recommendations

- Prioritise a root-cause review for haircare SKUs with higher defect-rate values.
- Investigate failed inspections by SKU, supplier, product type and manufacturing batch where batch data is available.
- Keep Pending records separate from Pass records until inspection is complete.
- Confirm the defect-rate definition, scale and acceptable threshold with the quality team.

The average defect-rate chart is descriptive. It does not identify the cause of defects or establish whether the differences are statistically meaningful.

9. Supplier & Manufacturing Performance

Supplier performance is compared using record count, average Lead time, average defect rate and average shipping cost. Manufacturing is compared by product type using average manufacturing lead time and manufacturing cost.

SUPPLIER	RECORDS	AVG. LEAD TIME	AVG. DEFECT RATE	AVG. SHIPPING COST
Supplier 1	27	14.78	1.80	5.51
Supplier 2	22	18.55	2.36	5.74
Supplier 3	15	20.13	2.47	4.79
Supplier 4	18	15.22	2.34	5.76
Supplier 5	18	18.06	2.67	5.79

Supplier observations

- Supplier 1 has the lowest average Lead time at 14.78 and lowest average defect rate at 1.80.
- Supplier 3 has the highest average Lead time at 20.13.
- Supplier 5 has the highest average defect rate at 2.67.

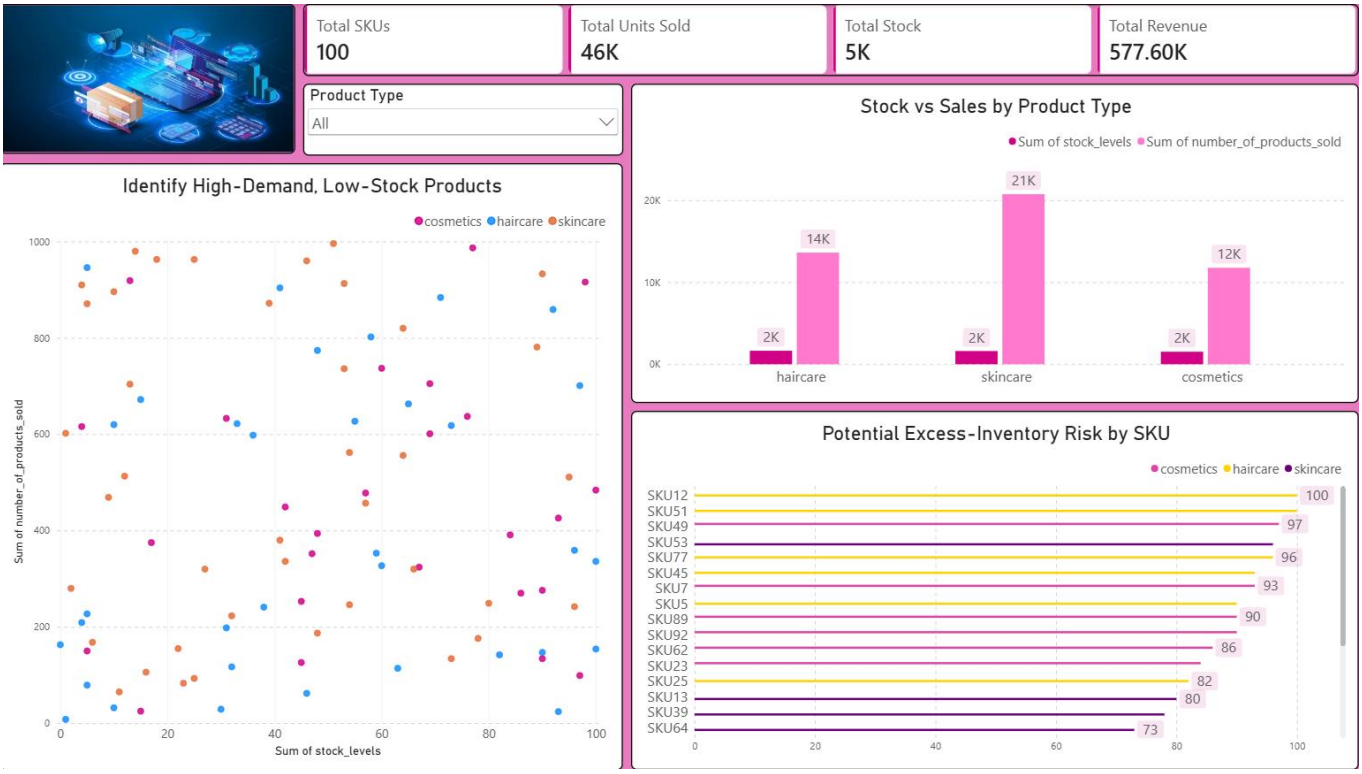
PRODUCT TYPE	AVG. MANUFACTURING LEAD TIME	AVG. MANUFACTURING COST
Haircare	17.06	48.46
Skincare	13.78	48.99
Cosmetics	13.31	43.05

Recommendations

- Discuss longer lead times and higher defect rates with relevant suppliers before changing supplier allocation.
- Compare suppliers within similar product types and locations to reduce product-mix bias.
- Track on-time delivery, incoming defects, rejected quantity and total landed cost in a future supplier scorecard.
- Confirm whether supplier lead time should use Lead time or Lead times before formal reporting.

10. Power BI Dashboard - Inventory Optimization Overview

This page provides a top-level view of SKU count, units sold, stock and revenue. It includes a product-type filter, a stock-versus-sales comparison, a SKU-level scatter plot and a list of SKUs with higher stock values for further review.



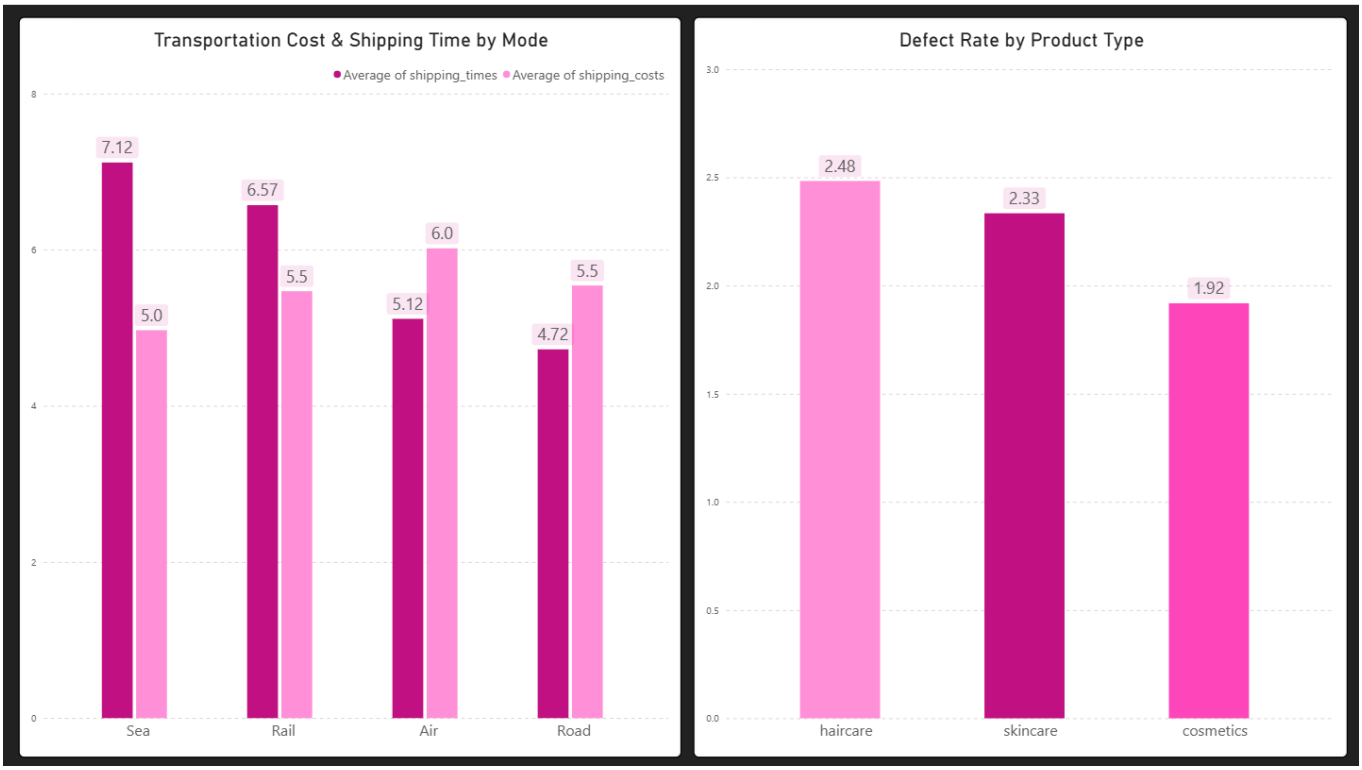
How to read this page

- KPI cards summarise the dataset totals.
- Stock vs Sales by Product Type compares stock with units sold across the three categories.
- The scatter plot helps identify products with relatively high sales and lower stock.
- The SKU stock chart is a review list; high stock alone does not prove excess inventory.

Business use: use this page to identify candidate SKUs for review, then validate inventory decisions against recent demand, replenishment time and business service-level targets.

11. Power BI Dashboard - Logistics & Quality Performance

This page compares transportation modes using shipping time and shipping cost, and compares product types using average defect rate. It supports a discussion about delivery speed, cost and product quality.



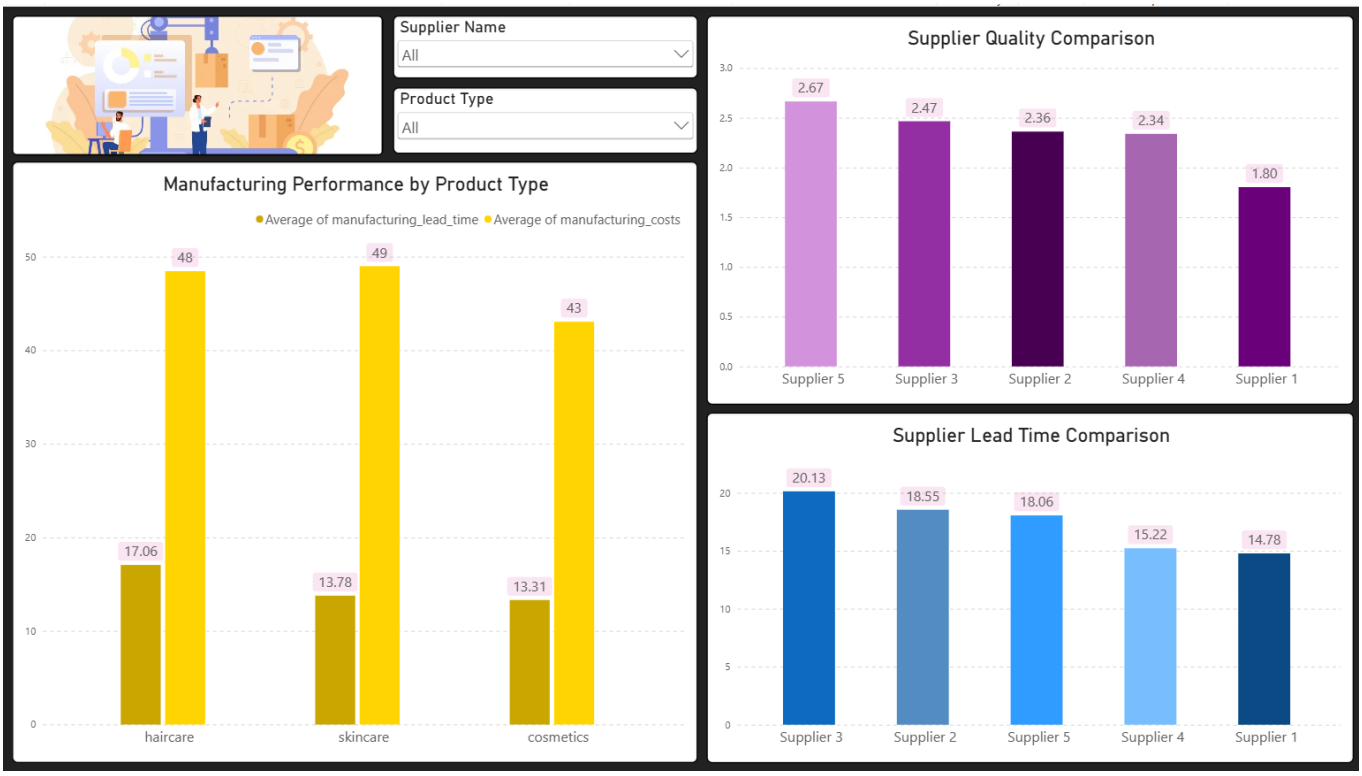
How to read this page

- The transportation chart compares average shipping time and cost for Sea, Rail, Air and Road.
- The defect-rate chart compares the average recorded defect rate for haircare, skincare and cosmetics.
- Use the visuals to identify where carrier, route or quality investigation may be useful.

Business use: use this page to identify candidate SKUs for review, then validate inventory decisions against recent demand, replenishment time and business service-level targets.

12. Power BI Dashboard - Supplier & Manufacturing Performance

This page compares supplier quality and lead time, and compares manufacturing lead time and manufacturing cost by product type. Supplier and product filters allow the report user to narrow the comparison.



How to read this page

- Supplier Quality Comparison shows average defect rate by supplier.
- Supplier Lead Time Comparison shows the average supplier Lead time field used in the report.
- Manufacturing Performance compares average manufacturing lead time and cost by product type.

Business use: use this page to identify candidate SKUs for review, then validate inventory decisions against recent demand, replenishment time and business service-level targets.

Recommendations, limitations & conclusion

Recommendations

- Review the 21 products that have high sales but low stock to check whether more stock is needed.
- Review the 25 products that have high stock but low sales before placing new orders.
- Compare suppliers based on delivery time and product quality.
- Check products that failed or are still waiting for quality inspection.
- Compare transportation methods based on delivery time and overall cost.
- Maintain extra stock for products with regular customer demand to avoid running out.

Limitations

- The dataset contains only 100 records.
- It does not include sales history or dates, so we cannot study changes in demand over time.
- Information about previous stock shortages, purchase orders and reorder levels is not available.
- Supplier agreements and detailed product quality records are not provided.

- Some details, such as measurement units and cost definitions, need confirmation.
- The analysis shows patterns in the available data but cannot predict future demand or explain the exact reasons behind the results.

Conclusion

In this project, I used PostgreSQL to analyse and check the supply chain data and Power BI to create interactive dashboards. The analysis helped identify products that may need more stock, compare transportation methods, understand product quality and review supplier performance.

Overall, this project helped me understand how data analysis can support better inventory management and supply chain decisions.